

Do Vision–Language Models Perceive Visual Illusions? A Systematic Evaluation of Human–AI Perceptual Misalignment

Han Jeong Seu, Caleb

1. BACKGROUND & RELATED WORK

1.1 VISUAL ILLUSIONS

Visual illusions happen when our brains interpret what we see in a way that doesn’t match the actual physical properties of the stimulus [1]. Scientists have studied these illusions for more than a hundred years, uncovering the mental shortcuts our visual system uses. These illusions can also be described by the types of perceptual judgments they influence—such as how we see contours or why still patterns sometimes seem to move.

1.2 VISUAL-LANGUAGE MODELS AND VISUAL GROUNDING

A Vision-Language Model (VLM) integrates Computer Vision (CV) and Natural Language Processing (NLP) to generate answers and explanations for image-text inputs. Such model competencies have been a key driving force in the healthcare sector, among others, where they demonstrate their ability to analyse medical images alongside text to mimic the cognitive processes of a medical professional [2]. The advent of VLMs may seem groundbreaking, especially given their ability to generate fluent, natural-sounding explanations. However, despite its linguistic plausibility, it’s not safe to assume that the answers generated are based solely on the image itself. There must be verification checks, which introduce visual grounding. Visual grounding refers to the process of linking textual outputs to specific regions of the visual input [3], ensuring that the model can substantiate its own responses with appropriate visual evidence.

1.3 RELATED WORK AND RESEARCH GAP

Earlier research focused on using visual illusions to understand how the human brain pieces together what we see, showing that humans rely on shortcuts and context to make sense of unclear information. These days, computer vision and vision-language models (VLMs) are judged mainly by how well they perform on tasks such as image classification, caption generation, and question answering [4]. While there are more tools now to help explain what models are doing, such as attention maps, these do not always show that the explanation is linked to the actual image (i.e. sometimes a model may sound right even if it isn’t). Most tests do not examine where models might go wrong in illusions, or require them to show precisely where their evidence comes from [5]. This study helps fill that gap by introducing a wide variety of illusions and a clear, step-by-step way to test whether GPT-4o’s answers are truly based on the images themselves.

1.4 HYPOTHESIS OF STUDY

This study hypothesises that GPT-4o will provide accurate text-based representations for the recognition and reasoning of illusions, respectively. However, its spatial grounding may be less reliable, especially for subtler variants of the illusion, where bounding boxes may become coarse or inconsistent. Beyond results, this study offers a viable course of action to enable increased reliability through more data-driven evaluation and correction workflows, emphasising visual evidence over clean phrasing. Taken together, this frames visual illusions as both a benchmark and a practical stress test for deciding when a model’s “understanding” should be trusted.

2. METHODOLOGY

To investigate the alignment between human perception and GPT-4o’s visual processing, we developed a multi-stage evaluation framework. This framework looks beyond simple image recognition, requiring the model to demonstrate understanding through perception, reasoning, and spatial grounding. Our method is shown in Figure 1 below.

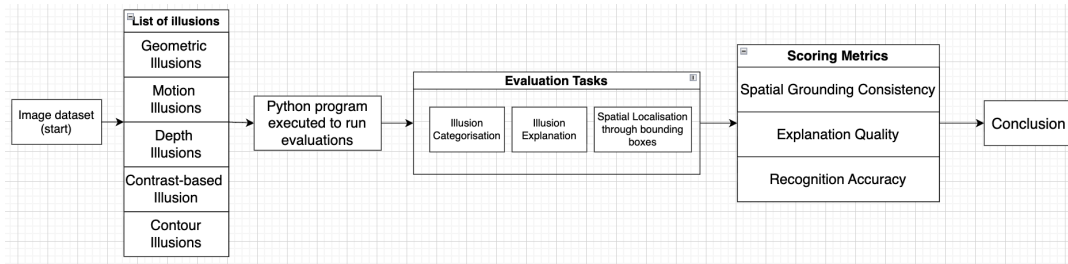


Figure 1

2.1 DATASET SUMMARY (✓)

A dataset of visual illusion images was selected to cover a broad range of illusion mechanisms. The dataset spans five categories: contour, contrast-based, depth, geometric, and motion. For each category, both “classic” examples and those with more subtle effects were used. These included reducing the risk that the model evaluates each image by recognising a set of conformed patterns, and instead testing whether the model’s outputs remain uniform across different variations of the exact illusion mechanism.

Illusion Type	Count	Examples (non-exhaustive)
Contour	100	Kanizsa triangle
Contrast-based	100	Brightness Induction
Depth	100	Ponzo Illusion
Geometric	100	Müller-Lyer
Motion	100	Waterfall Illusion

Fig 1.1

2.2 EVALUATION TASKS

Each experiment in this study was purposefully tailored to guide the tasks GPT-4o performed across all illusion images and trials. The prompts were designed to evaluate the individual model’s capabilities (i.e., recognition, reasoning, and image localisation).

1. Categorisation Prompt

- Posed the question to the system to categorise the data into exactly one illusion mechanism from the list: Contour, Contrast-based, Depth, Geometric and Motion illusions.
- To evaluate the capability of GPT-4o to associate an image with the actual perceptual cue, instead of similarity-based matching with other images.

2. Explanation prompt

- Generate one question related to the visual illusion, and based on that question, give an answer that relates to the specific features of the image (visual cues, etc.).
- Consider whether the generated justification includes links back to the image rather than a generic response.

3. Localisation Prompt

- Localise illusion-inducing area(s) with pixel bounding box (x1, y1, x2, y2).
- Test for visual grounding by making a concrete evidence placement.

The prompts specific to each evaluation are located in Annexe A, under A.1, A.2 and A.3, respectively.

2.3 SCORING CRITERIA AND METRICS

For the categorisation tasks, the predicted label for each image was directly compared to the ground-truth label in the dataset, and performance was evaluated as percentage accuracy across all N images.

$$Accuracy (\%) = \frac{No. \text{ of Correct Predictions}}{N} \times 100$$

For the explanation task, GPT-4o provided a brief justification for why it categorised the image under the chosen category. A manual assignment was made to categorise them using a rubric label. Grounded responses typically reference visible cues in the image; generic responses were linguistically plausible but required image-specific references, while incorrect responses had no relation to the image and mismatched the visual mechanisms shown.

$$p_k = \frac{1}{N} \sum_{i=1}^N (r_i = k), \quad k \in \{Grounded, Generic, Incorrect\}$$

For the localisation task, GPT-4o was tasked to output pixel-coordinates of bounding box(s) that localise the coordinates of the illusion-inducing area(s) of the image. The output was parsed and recorded in a CSV file, along with a validity label. These labels were assigned based on how well the predicted box captured the illusion-inducing region. Accurate responses generally referenced all the illusion-inducing areas in the image, with accurate localisation; Oversized responses generally referenced the entire image or the majority of the image without explicitly localising onto the specific illusion-inducing area(s); Partial responses generally referenced part of the illusion, cutting out halfway through the illusion-inducing areas; Invalid responses were either entirely off the illusion-inducing image, or had no output generated at all.

$$q_m = \frac{1}{N} \sum_{i=1}^N (l_i = m), \quad m \in \{Accurate, Oversized, Partial, Invalid\}$$

Overall, these scoring criteria allow us to quantify our categorisation, explanation and localisation outputs, making it easier for further evaluation upon the completion of the experiment.

2.4 EXPERIMENTAL SETUP

2.4.1 IMPLEMENTATION AND CODE ARCHITECTURE

Implementations were performed on OpenAI API (GPT-4o). The analysis pipeline was developed in Python 3.13.2 and implemented in Visual Studio Code, with version control managed with GitHub.

Model: GPT-4o (Version: gpt-4o-2024-11-20)

Libraries:

1. OpenAI: For API interaction and prompt handling.
2. Pillow (PIL) & OpenCV: For image processing and bounding box visualisation.
3. Pandas & NumPy: For data management and statistical analysis.
4. Matplotlib & Seaborn: For generating performance visualisations.

3. RESULTS & DISCUSSION

3.1 CATEGORISATION PERFORMANCE

The categorisation performance was evaluated by comparing GPT-4o’s output predicted mechanism category against its ground-truth mechanism label. Based on the experiments, this study concludes that GPT-4o’s ability to categorise different mechanisms is reliable at a broad level; however, it is inconsistent across mechanisms, as some may share overlapping visual cues, specifically depth and geometric illusions. The full categorisation outputs are provided in Annexe B, Table B1. The question remains whether the model associates an image with the actual perceptual cue rather than performing similarity-based matching [6].

3.2 EXPLANATION PERFORMANCE

The quality of explanations was evaluated based on grounding. Figure 3 shows the distribution of grounded, generic, and incorrect responses. There were numerous instances where, despite GPT-4o’s ability to categorise mechanisms and produce coherent explanations, it still struggled to ground claims to precise visual evidence. The model occasionally answers based purely on how one perceives them to look. At times, the model also incorrectly described features of the image itself, answering questions generated by the system incorrectly. Overall, despite the model’s linguistic fluency in generating both questions and answers, it still lacked proper visual grounding for the image, and its claims were not properly anchored to the specific visual evidence.

3.3 LOCALISATION PERFORMANCE

Generally, GPT seemed accurate in identifying simple illusions, as shown in Figure 3.1 (Annexe C), where a red bounding box outlines the portion of an image that contains the illusion of a moving circle pattern. In Figure 3.2 (Annexe C), there were instances where GPT seemed completely inaccurate, where it outlined the spaces between the two yellow lines, which had no illusion-inducing region at all. Instead, the expected localisation area should have outlined both the top and bottom yellow lines, outlining the relevant components of the Müller-Lyer illusion. The prompt stated that GPT should outline multiple bounding boxes if the same instance of an illusion appears a few times, and prevent drawing boxes over the entire image unless the entire frame itself forms an illusion. Despite the constraint mentioned, GPT still frequently collapsed multiple illusion-inducing areas into a single box. An attempt was made to split the bounding boxes into individualised illusion components, but GPT still failed to identify all the illusions in the image, as shown in Figure 3.3 (Annexe C). There were also instances in which GPT drew a single bounding box over the entire image, even though the image contained numerous sub-regions with illusion-inducing areas, as shown in Figure 3.4 (Annexe C). Overall, GPT isn’t very accurate at identifying depth illusions. These illusions range from the Ponzo Illusion (where two identical lines are perceived as different lengths) to the Size Constancy Illusion. However, GPT is accurate when identifying motion illusions, mostly the tunnel depth illusion.

4. DISCUSSION

4.1 Key takeaway: Visual grounding vs Linguistic Fluency

There were numerous instances where, despite GPT-4o’s ability to categorise mechanisms and produce coherent explanations, it still struggled to ground claims to precise visual evidence. There were two aspects of the experiment that demonstrated this. Firstly, the model outputs generic explanations, and secondly, the model provides a coarse localisation (e.g. localising the entire image, or specific elements not relating to illusion-induced regions)

4.2 Why certain categories may fail

For certain categories, they exhibit similar physical traits or may have some subtle overlap.

- Depth and Geometric illusions, which both rely on perspective and converging lines, blur the boundaries of mechanism.
- Contour and Contrast Illusions, where luminance context may co-occur, causing confusion
- Motion Illusions that may require intricate spatial interpretation before illusion is seen which makes visual grounding more difficult to achieve.

4.3 Our proposed workaround

This study aims to propose a Evidence-Constrained Illusion Evaluation (ECIE) workflow.

4.3.1 Two-stage prompting

This workflow captures the idea of sequential dependency between localisation and reasoning.

1. GPT-4o should produce a single bounding box around the region most concentrated by illusion-inducing effects.
2. Upon producing it, GPT-4o should provide a short justification as to why it explicitly localised this spatial subset in particular, highlighting the visual cues which prompted it to localise this region

Through this restriction, it reduces the models reliance on pre-trained descriptions, which forces the model to think independently without external context. This allows us too have a better understanding of the models capabilities to capture mechanism-level reasoning from evidence that has strong visual grounding.

5. LIMITATIONS & ETHICAL CONSIDERATIONS

5.1 LIMITATIONS

This study focuses on only one type of vision-language model, GPT-4o. They can analyse GPT-4o closely, but they cannot confirm whether their findings apply to all vision-language models. This is because different models are trained on different data and constructed in different ways, so their responses to visual illusions will be different. The dataset features many types of illusions organised into five main groups, but it is not exhaustive. Certain types of illusions, such as hybrid illusions and dynamic perceptual effects, are not included. This means the dataset omits some well-known illusions. Finally, the study relies on fixed, careful prompts. While this provides consistency, it will definitely affect the performance.

5.2 ETHICAL CONSIDERATIONS

This study does not involve the collection of live vertebrae, sensitive or personal data. All visual stimuli used were sourced from publicly available, open-access datasets (e.g., Hugging Face) to ensure ethical research standards.

6. CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This study examined how VLMs perceive illusions. GPT-4o did recognise illusions that involve the features such as edges of things, their brightness, and how they move. However, it had a hard time telling how deep things are, what shape they have and even coming up with explanations that weren't about what was actually in the picture. When it tried to locate objects in the picture, it often got them wrong, showing that there is a difference between being able to come up with answers and truly understanding what it is seeing.

6.2 FUTURE WORK

Future research could explore the synthetically generated illusion datasets and evaluate a wider range of model architectures while integrating quantitative localisation metrics with human benchmark comparisons so as to rule more of these illusions. With stronger integration between visual grounding and explanation generation, VLMs will further enhanced in terms of the reliability and interpretability.

7. REFERENCES

The author declares that Generative AI was used in this project. I recognise the use of GenerativeAI in this report as part of the experimental pipeline, specifically the usage of the GPT-4o visual-language model. This visual-language model was used to assist with experimental procedures, namely the image classification, explanation, and localisation of visual illusions in the input image. The use of GenerativeAI is strictly limited to the above methods and is not used elsewhere in this report.

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9. APPENDIX

ANNEXE A

A1 CATEGORISATION PROMPT

You will be shown an image that produces a known visual illusion in human perception.
 From the list below, identify the primary perceptual mechanism responsible for the illusion.
 Choose from one of the categories below only

- Contour
- Contrast-based
- Depth
- Geometric
- Motion

Respond using only the category name, with no additional explanation.

A2 EXPLANATION PROMPT

You will be shown an image that produces a known visual illusion in human perception.
 Before generating a description of the illusion, you are to state exactly one question that captures this perceptual judgement (e.g. “Which line appears longer?” or “Which square appears bigger?”).
 Answer the question you generated, and explain why the illusion occurs
 Your response should:

- Contain exactly one question followed immediately by its answer
- Target a specific perceptual property visible in the image (e.g. length, brightness, depth, alignment)
- Refer only to the visual evidence present in the image
- Avoid generic or abstract explanations

A3 LOCALISATION PROMPT

You will be shown an image that produces a known visual illusion in human perception.
 Your task is to identify the region(s) in the image most responsible for producing the illusion.
 If multiple illusion-inducing areas exist, output multiple entries in the ‘illusions’ list, and make sure to keep the bounding boxes tight around the key illusion-inducing area.
 Your response should enable the reader to visually locate the relevant illusion-inducing area in the image by outputting a set of pixel coordinates to localise the illusion.

ANNEXE B

TABLE B1

Illusion Mechanism	N (images)	Correct	Accuracy (%)
Contour	100	100	100
Contrast-based	100	100	100
Depth	100	48	48
Geometric	100	57	57
Motion	100	79	79
Total	500	1,384	76.8

ANNEXE C

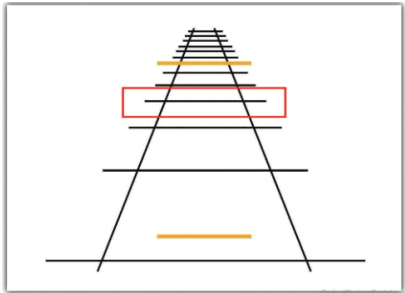


Fig 3.1

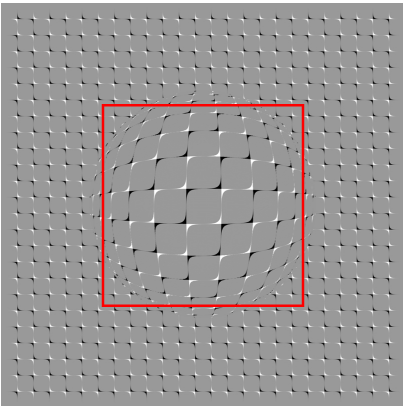


Fig 3.2

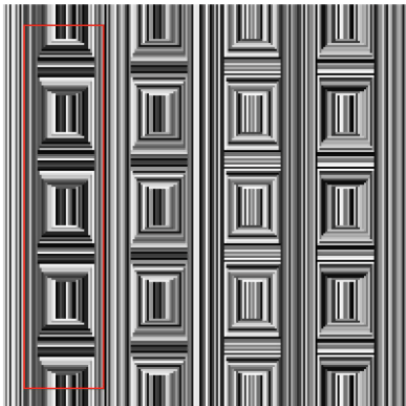


Fig 3.3

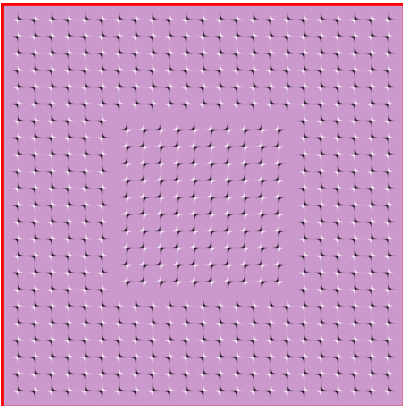


Fig 3.4